Lecture Notes: The Central Limit Theorem (CLT)

I. Introduction: The Core Concept

The **Central Limit Theorem (CLT)** is one of the most important and fundamental concepts in statistics.

It does not describe the distribution of individual data points. Instead, it describes the **distribution of *sample means***.

II. A Demonstration: The Magical Die

1. The Population Distribution

Imagine we have a single, magical 6-sided die. If we roll it thousands of times, the probability of rolling a 1, 2, 3, 4, 5, or 6 is equal (1/6).

This "population" of all possible rolls has a **Uniform Distribution**. It looks flat, not like a bell curve.

2. The Distribution of Sample Means (n=2)

Now, what if we roll the die **twice** ($n = 2$) and calculate the **average** of those two rolls? Let's look at the distribution of these *averages*.

- There's only one way to get an average of 1: rolling a (1, 1). The probability is $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.
- There's only one way to get an average of 6: rolling a (6, 6). The probability is $\frac{1}{36}$.
- But there are *many* ways to get an average of 3.5: (1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3). The probability is $\frac{6}{36}$.

The distribution of these *averages* is no longer flat. It's triangular, with the middle values being most common.

3. The Distribution of Sample Means (n=3)

What if we roll the die **three times** ($n = 3$) and take the average?

- Getting an average of 1 is now even rarer: (1, 1, 1) or $\frac{1}{216}$.
- The probabilities for the middle values (like 3.5) become even higher.

The shape of this new distribution (the distribution of the averages) starts to look very much like a **Normal Distribution**.

4. The Conclusion of the Experiment

As we increase our sample size (n)—rolling 10 times, or 20 times, and taking the average—the distribution of those sample averages becomes more and more perfectly normal.

III. Defining the Central Limit Theorem

This experiment demonstrates the theorem. The CLT states that:

> If we take a bunch of samples from a population and calculate the **mean** of each sample, the **distribution of those sample means** will be a **Normal Distribution**.

This is true **regardless of the shape of the original population distribution**. The population can be uniform (like the die), bimodal, or any "wacky" shape.

IV. The Three Key Results of the CLT

When the CLT is in effect, the resulting distribution of sample means has three specific properties:

1. **Shape:** The distribution of sample means will be a **Normal Distribution**.
2. **Center:** The **mean of the distribution of sample means** will be **equal to the original population mean (μ)**.
3. **Spread:** The **standard deviation of the distribution of sample means** will be **smaller** than the population standard deviation. It is precisely:

$$\frac{\sigma}{\sqrt{n}}$$

Where σ is the original population standard deviation and n is the sample size.

This special standard deviation (the one for the sample means) is called the **Standard Error of the Mean (SEM)**.

V. Second Example: Bimodal Mouse Weights

Let's test this on a different population.

- **Population:** Imagine a "wacky" bimodal distribution of mouse weights (e.g., a mix of two different species, one small and one large). This distribution is *not* normal.
- **Process:**
    i. We take a sample of $n = 2$ mice and calculate their average weight.
    ii. We take another sample of $n = 2$ and get another average.
    iii. We repeat this 1000 times.
- **The Result:**
    - The histogram of our 1,000 *sample means* looks like a **Normal Distribution**, even though the original population was bimodal.
    - The mean of this new distribution is centered on the original population mean.
    - The standard deviation (SEM) of this new distribution is equal to the population standard deviation divided by $\sqrt{2}$.

VI. How "Large" is "Large Enough"?

How big does our sample size (n) need to be for the CLT to apply?

- **If the original population is *already* normal:** The CLT applies for *any* sample size. A sample size of $n = 2$ is large enough.
- **If the original population is "wacky":** A sample size of $n = 30$ **is often considered the rule of thumb** for the CLT to apply.

VII. Why is the CLT So Important?

This is the key takeaway.

Many of our most common and powerful statistical tests (like t-tests, Z-tests, ANOVA) were all designed to work on data that is **normally distributed**.

But what if our *population* data is not normal (like the bimodal mice)? Does that mean we can't use these tests?

The **CLT saves us!** It says that even if our population data is not normal, the **distribution of the *sample means* IS normal** (as long as our sample size n is large enough, e.g., > 30).

Because the sample means are normally distributed, we can confidently use t-tests and other normal-based tests on our data, and this is what makes statistics so powerful.